Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data into Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks Cluster
- Databricks Runtime Version: 11.3.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: customerdata
- Location: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: orderdata
- Location: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: customer
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Customer data in Delta format
- Output Name: order

- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages: Read CSV data, create Delta tables, load data into Delta tables

Data Transformations / Business Logic:

- Step: Read customer CSV data
- Description: Read customer CSV data into a DataFrame
- Transformation Logic: Use `spark.read.csv` to read CSV data
- Step: Read order CSV data
- Description: Read order CSV data into a DataFrame
- Transformation Logic: Use `spark.read.csv` to read CSV data

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-002: Transform Order Data by Adding TotalAmount Column

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a Delta Live Table to add a new column "TotalAmount" to the "order" Delta table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: order
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: order
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Order data with TotalAmount column in Delta format

Job Flow / Pipeline Stages: Read order Delta table, add TotalAmount column, write updated data to Delta table

Data Transformations / Business Logic:

- Step: Add TotalAmount column
- Description: Add a new column "TotalAmount" to the "order" Delta table

- Transformation Logic: Use `withColumn` to add a new column with the product of " PricePerUnit" and "Qty" columns

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-003: Cleanse Customer and Order Data

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table to remove null and duplicate records from "customer" and "order" Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: order
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: customer
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Cleansed customer data in Delta format
- Output Name: order
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Cleansed order data in Delta format

Job Flow / Pipeline Stages: Read customer and order Delta tables, remove null and duplicate records, write updated data to Delta tables

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove null records from customer and order DataFrames
- Transformation Logic: Use `dropna` to remove null records
- Step: Remove duplicate records

- Description: Remove duplicate records from customer and order DataFrames
- Transformation Logic: Use `dropDuplicates` to remove duplicate records

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-004: Create ordersummary Table and Load Joined Data

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a Delta Live Table to create an "ordersummary" table and load joined data from "customer" and "order" Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: order
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: ordersummary
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Joined customer and order data in Delta format

Job Flow / Pipeline Stages: Read customer and order Delta tables, join data, create ordersummary table, load joined data into ordersummary table

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join customer and order DataFrames on CustId
- Transformation Logic: Use `join` to join DataFrames

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-005: Implement SCD Type 2 Logic for ordersummary Table

Related Functional Requirement(s): FR-DLT-005

Objective: Implement SCD Type 2 logic to update the "ordersummary" table when there are changes in the "customer" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Customer data in Delta format

Target Data Details:

- Output Name: ordersummary
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Updated ordersummary data in Delta format

Job Flow / Pipeline Stages: Read customer Delta table, detect changes, update ordersummary table accordingly

Data Transformations / Business Logic:

- Step: Detect changes in customer data
- Description: Detect changes in customer DataFrame
- Transformation Logic: Use `exceptAll` to detect changes
- Step: Update ordersummary table
- Description: Update ordersummary table with changed customer data
- Transformation Logic: Use `merge` to update ordersummary table

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Related Functional Requirement(s): FR-DLT-006

Objective: Implement a Delta Live Table to create a "customeraggregatespend" table and load aggregated data from the "ordersummary" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary

- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Joined customer and order data in Delta format

Target Data Details:

- Output Name: customeraggregatespend
- Location: Delta table in `gen_ai_poc_databrickscoe.sdlc_wizard` schema
- Format: Delta
- Description: Aggregated customer spend data in Delta format

Job Flow / Pipeline Stages: Read ordersummary Delta table, aggregate data, create customeraggregatespend table, load aggregated data into customeraggregatespend table

Data Transformations / Business Logic:

- Step: Aggregate data
- Description: Aggregate TotalAmount column by Name and Date
- Transformation Logic: Use `groupBy` and `sum` to aggregate data

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms

TR-DLT-007: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-007

Objective: Implement Delta Live Tables to automate the execution of the above steps.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages: Create Delta Live Tables pipeline, add above steps to pipeline, execute pipeline

Error Handling and Logging: Log errors and exceptions using Databricks logging mechanisms